Filter for the ADC/DAC

Caps for STM32 supply

SWD connector

I2C SSD1306 Display

UART CONNECTOR

CAN HEADER

Static discharge USB protection

Rotary encoder/switch
with a debouncing circuit

Resonator

Debug LED indicators

TIM4 ENC MODE

SYS power
indicators

Power to the
on-board computer

Motor driver 0

File: motor_driver.kicad_sch

Motor driver 1

File: motor_driver.kicad_sch

Motor driver 2

File: motor_driver.kicad_sch

Addressable LEDs around the PCB

File: leds.kicad_sch

External devices

File: ext_devices.kicad_sch

File: power_mgmnt.kicad_sch

File: dc_dc_ext.kicad_sch

File: dc_dc_12v_motors.kicad_sch

Motor wiring:
Pin | Color | Function
1 Red: Motor power
2 Black: Motor power
3 Green: Encoder ground
4 Blue: Encoder Vcc (3.5 V to 20 V)
5 Yellow: Encoder A output
6 White: Encoder B output

Mounting holes
(isolated)

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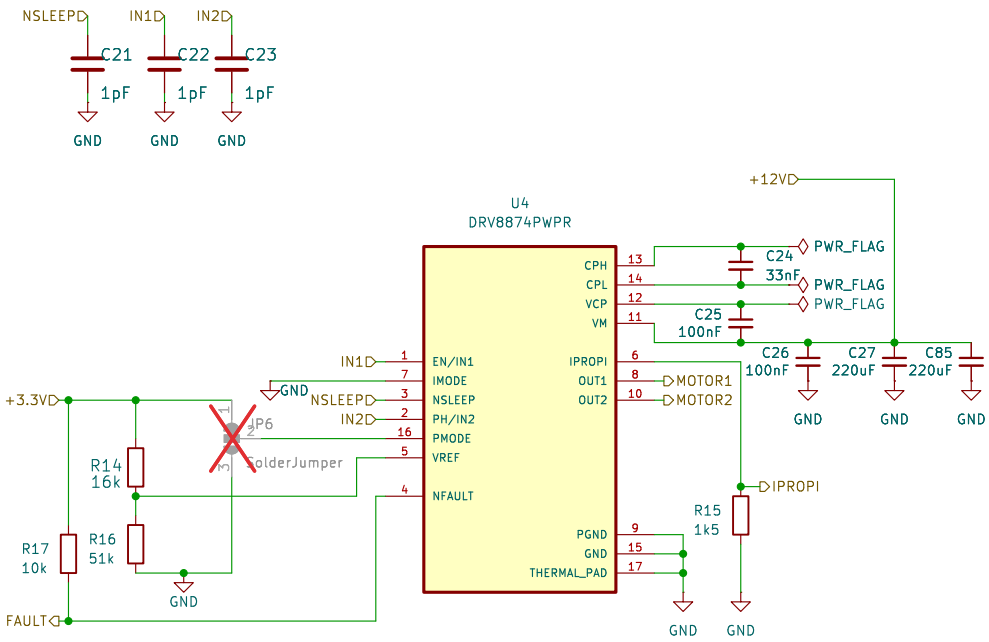
University of Tartu

Sheet: /

File: mainboard.kicad_sch

Title: Robotont mainboard

Size: A3	Date: 2023-21-11	Rev: 0.2.0
KiCad E.D.A. 9.0.6		Id: 1/18



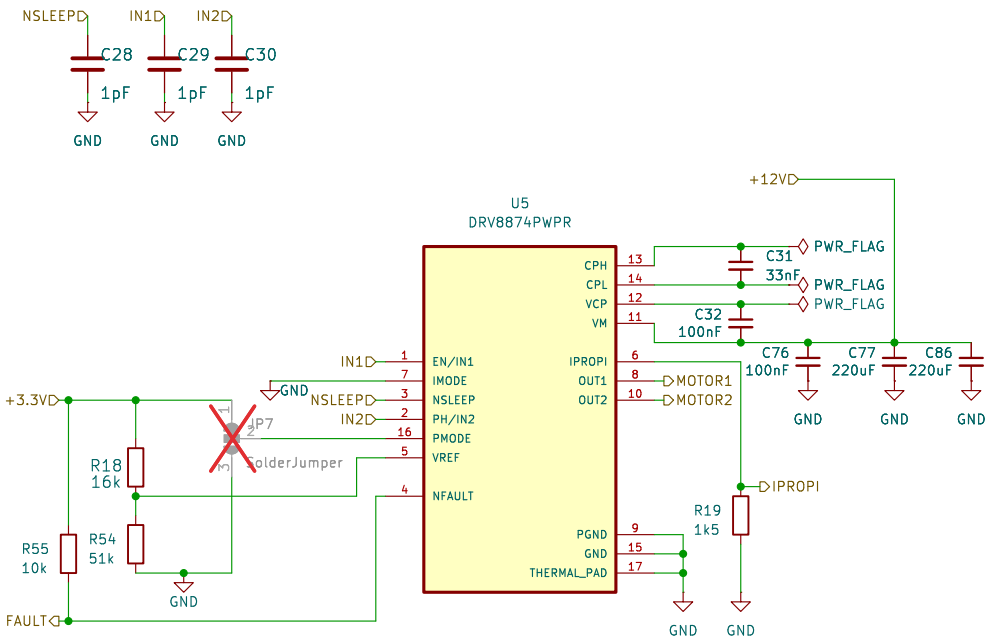
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Sheet: /Motor driver 0/
File: motor_driver.kicad_sch

Title: Motor driver

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 2/18



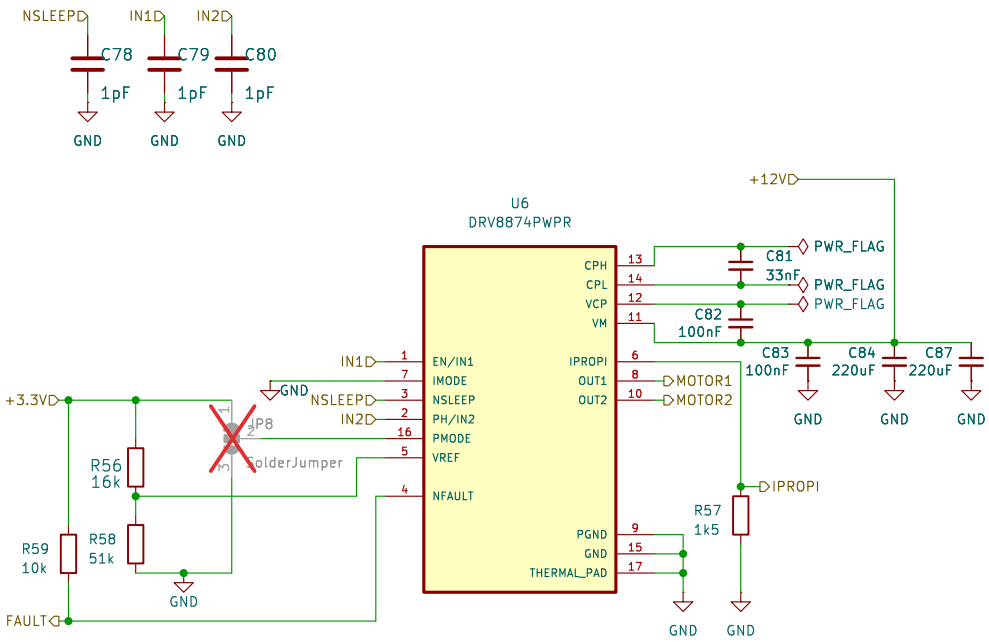
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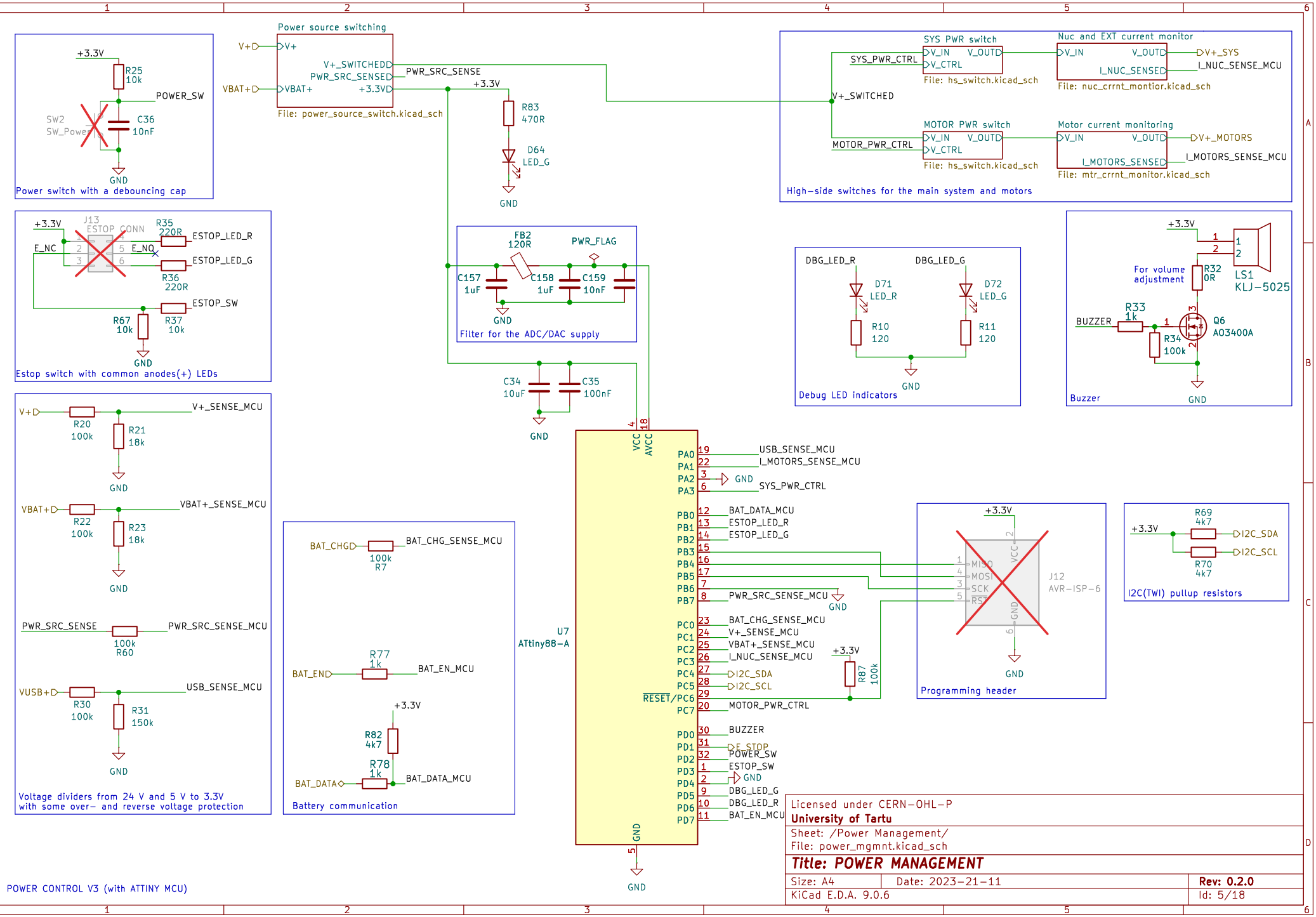
Sheet: /Motor driver 1/
File: motor_driver.kicad_sch

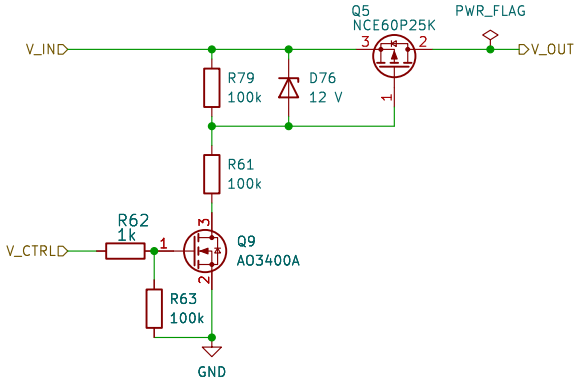
Title: Motor driver

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 3/18



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Sheet: /Motor driver 2/		
File: motor_driver.kicad_sch		
Title: Motor driver		
Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 4/18


Power switch with a debouncing cap
+3.3V
R25
10k
POWER_SW
SW2
SW_Power
C36
10nF
GND
Power source switching
V+
VBAT+
V+_SWITCHED
PWR_SRC_SENSE
+3.3V
File: power_source_switch.kicad_sch
R83
470R
D64
LED_G
GND
SYS PWR switch
SYS_PWR_CTRL
V_IN
V_OUT
V_CTRL
File: hs_switch.kicad_sch
Nuc and EXT current monitor
V+_SYS
I_NUC_SENSED
I_NUC_SENSE_MCU
File: nuc_crrnt_montior.kicad_sch
MOTOR PWR switch
MOTOR_PWR_CTRL
Motor current monitoring
V+_MOTORS
I_MOTORS_SENSED
I_MOTORS_SENSE_MCU
File: mtr_crrnt_monitor.kicad_sch
High-side switches for the main system and motors
Estop switch with common anodes(+) LEDs
J13
ESTOP_CONN
E_NC
E_NO
R35
220R
ESTOP_LED_R
ESTOP_LED_G
R36
220R
ESTOP_SW
R67
10k
R37
10k
FB2
120R
PWR_FLAG
C157
1uF
C158
1uF
C159
10nF
Filter for the ADC/DAC supply
C34
10uF
C35
100nF
DBG_LED_R
DBG_LED_G
D71
LED_R
D72
LED_G
R10
120
R11
120
Debug LED indicators
For volume adjustment
R32
0R
LS1
KLJ-5025
BUZZER
R33
1k
Q6
AO3400A
R34
100k
Buzzer
Voltage dividers from 24 V and 5 V to 3.3V
with some over- and reverse voltage protection
V+
R20
100k
R21
18k
V+_SENSE_MCU
VBAT+
R22
100k
R23
18k
VBAT+_SENSE_MCU
PWR_SRC_SENSE
100k
R60
PWR_SRC_SENSE_MCU
VUSB+
R30
100k
R31
150k
USB_SENSE_MCU
Battery communication
BAT_CHGD
100k
R7
BAT_CHG_SENSE_MCU
BAT_EN
R77
1k
BAT_EN_MCU
R82
4k7
R78
1k
BAT_DATA
BAT_DATA_MCU
U7
ATtiny88-A
VCC
AVCC
PA0
PA1
PA2
PA3
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PC0
PC1
PC2
PC3
PC4
PC5
RESET/PC6
PC7
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
USB_SENSE_MCU
I_MOTORS_SENSE_MCU
SYS_PWR_CTRL
BAT_DATA_MCU
ESTOP_LED_R
ESTOP_LED_G
PWR_SRC_SENSE_MCU
BAT_CHG_SENSE_MCU
V+_SENSE_MCU
VBAT+_SENSE_MCU
I_NUC_SENSE_MCU
I2C_SDA
I2C_SCL
MOTOR_PWR_CTRL
BUZZER
E_STOP
POWER_SW
ESTOP_SW
DBG_LED_G
DBG_LED_R
BAT_EN_MCU
R87
100k
Programming header
J12
AVR-ISP-6
MISO
MOSI
SCK
RST
I2C(TWI) pullup resistors
R69
4k7
R70
4k7
I2C_SDA
I2C_SCL
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University of Tartu
Sheet: /Power Management/
File: power_mgmnt.kicad_sch
Title: POWER MANAGEMENT
Size: A4
Date: 2023-21-11
Rev: 0.2.0
KiCad E.D.A. 9.0.6
Id: 5/18
POWER CONTROL V3 (with ATTINY MCU)



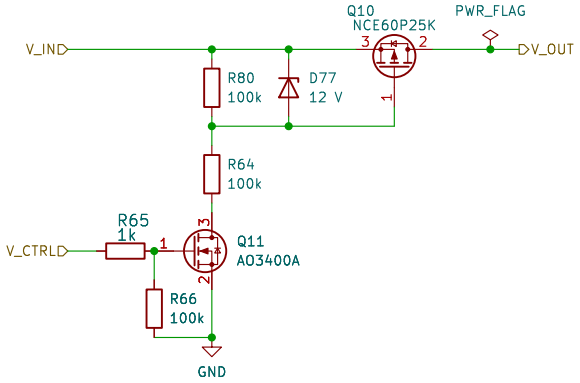
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Sheet: /Power Management/SYS PWR switch/
File: hs_switch.kicad_sch

Title: High side switch

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 14/18



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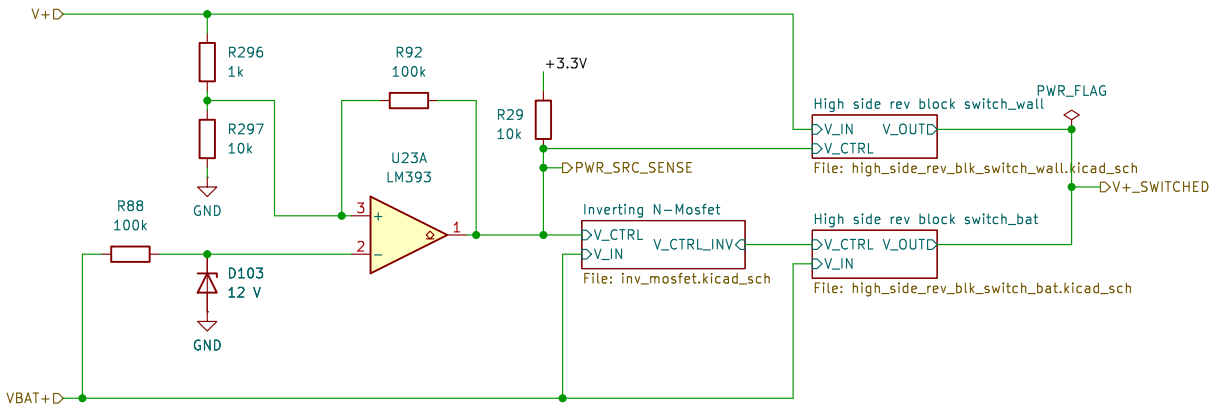
University of Tartu

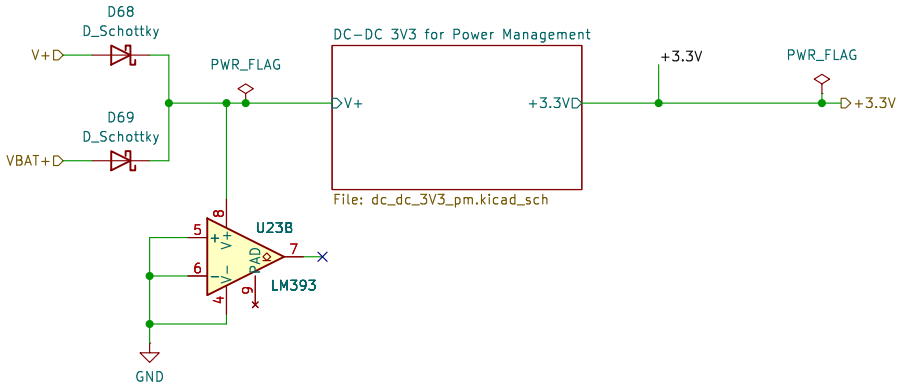
Sheet: /Power Management/MOTOR PWR switch/
File: hs_switch.kicad_sch

Title: High side switch

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 15/18

Zener defines reference voltage.
V+ will be used as the main power input once its voltage level on the voltage divider exceeds the zener brekdown threshold.





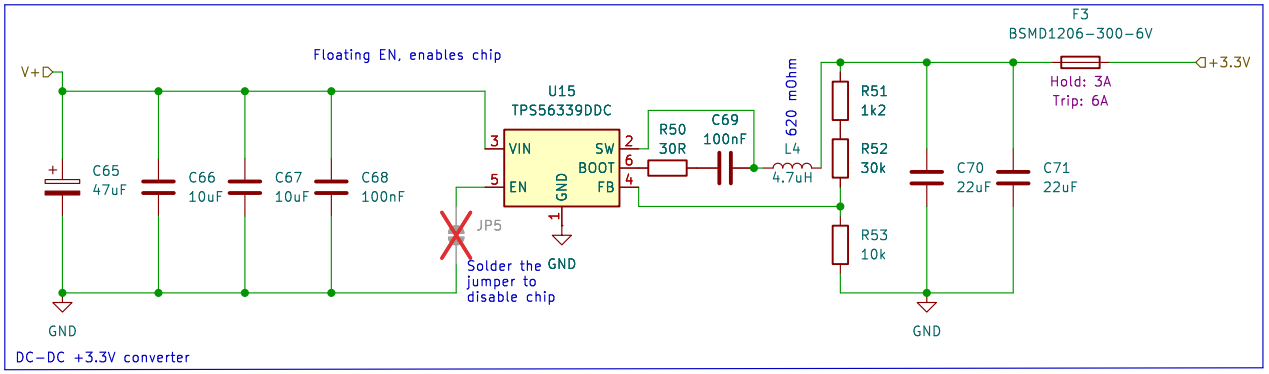
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Sheet: /Power Management/Power source switching/
File: power_source_switch.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 16/18

Required current max ~200mA



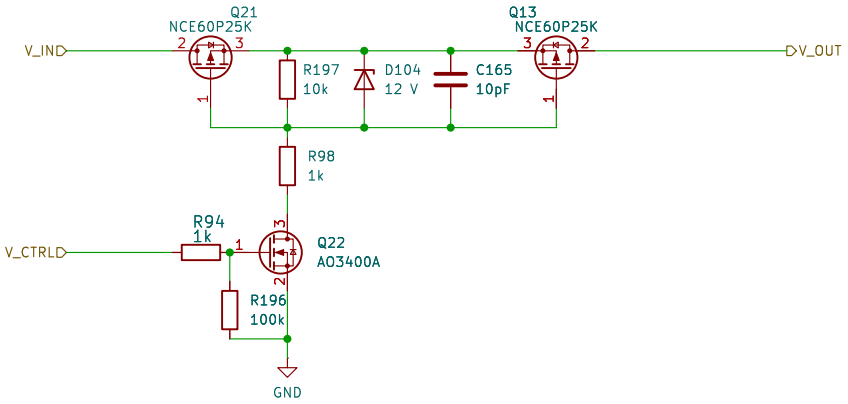
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Sheet: /Power Management/Power source switching/DC-DC 3V3 for Power Management/
File: dc_dc_3V3_pm.kicad_sch

Title: DC-DC converter for the power management subsystem

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 6/18

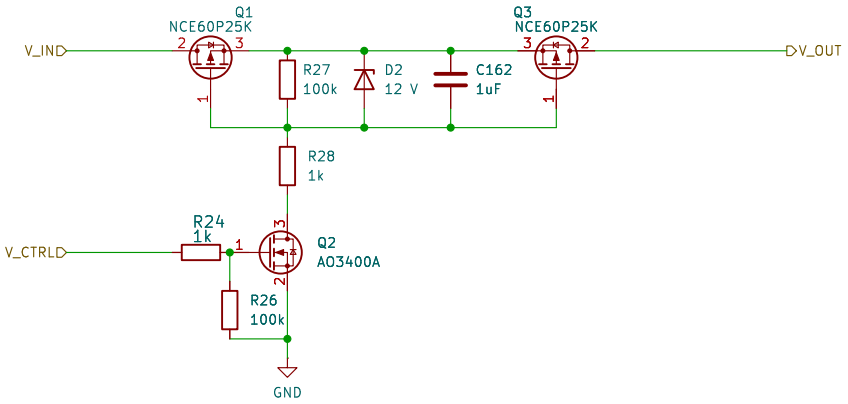


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Sheet: /Power Management/Power source switching/High side rev block switch_wall/
File: high_side_rev_blk_switch_wall.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 11/18

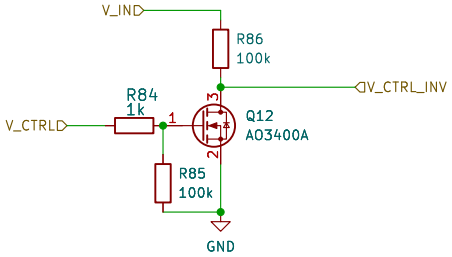


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Sheet: /Power Management/Power source switching/High side rev block switch_bat/
File: high_side_rev_blk_switch_bat.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 12/18

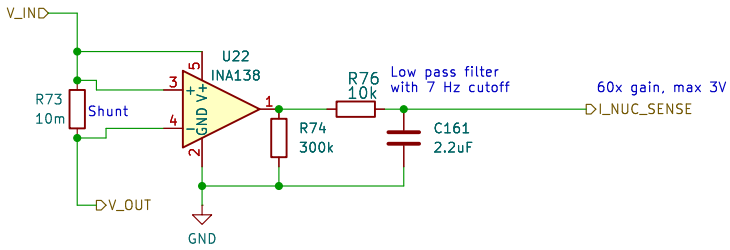


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Sheet: /Power Management/Power source switching/Inverting N-Mosfet/
File: inv_mosfet.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 18/18



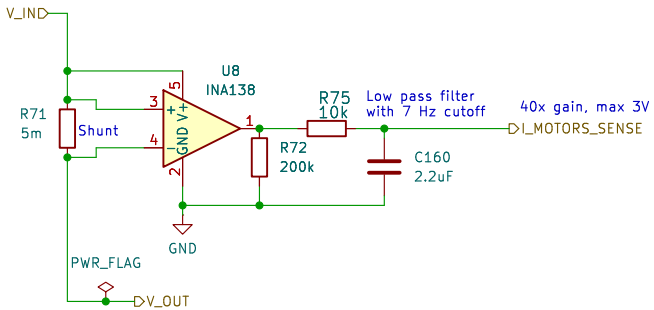
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Sheet: /Power Management/Nuc and EXT current monitor/
File: nuc_crrnt_montior.kicad_sch

Title: Onboard computer current sensing circuit

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 17/18



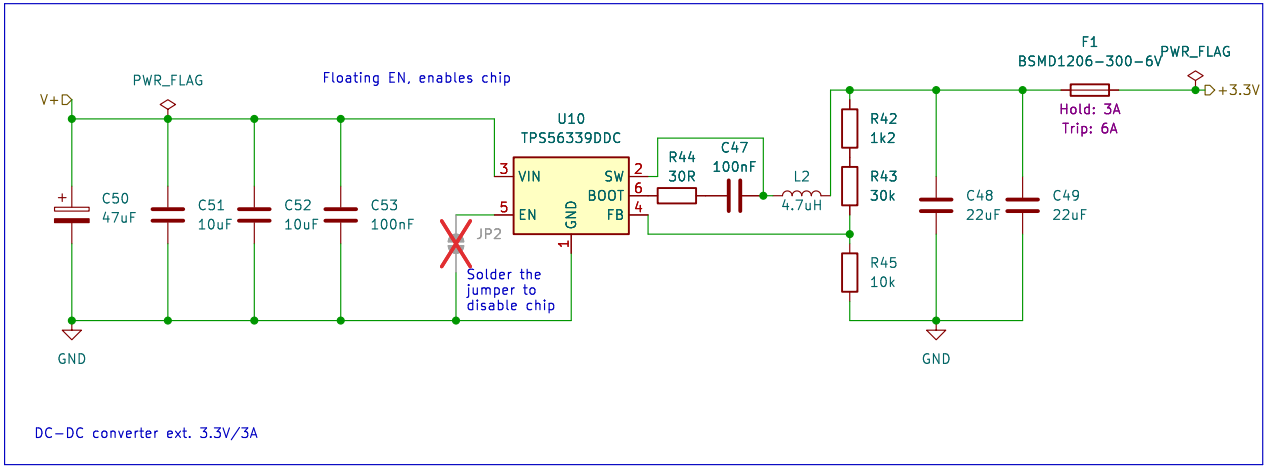
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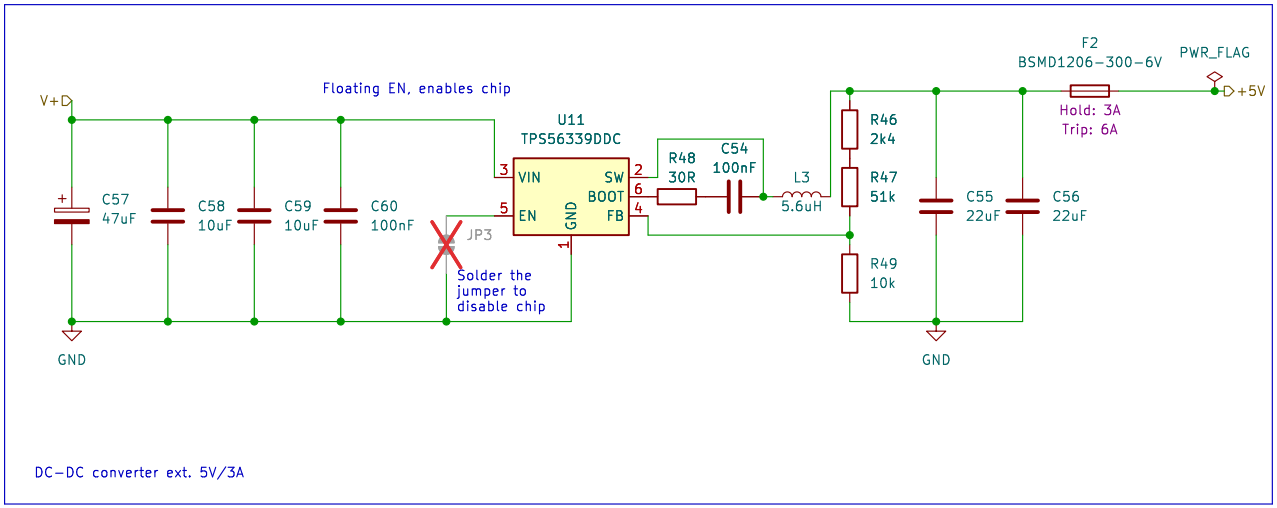
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Sheet: /Power Management/Motor current monitoring/
File: mtr_crrnt_monitor.kicad_sch

Title: Motor current sensing circuit

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 17/18





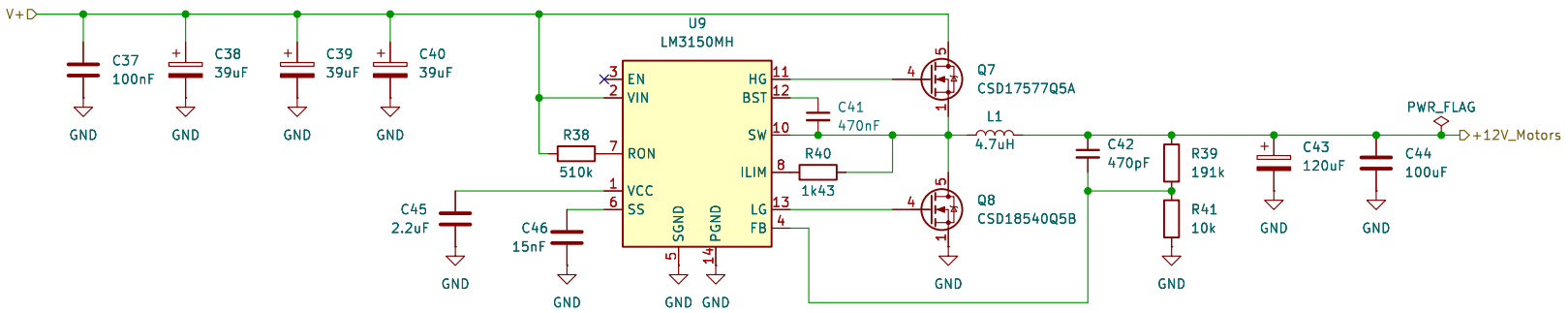
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Sheet: /DC-DC for EXT, MCU, and LEDs/
File: dc_dc_ext.kicad_sch

Title: DC-DC converter for the MCU, LEDs, and external devices

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 7/18



R_on = V_out / (K*f),
where K=100 pF is constant.
For R_on = 510 kOhm, fs ~= 235 kHz.

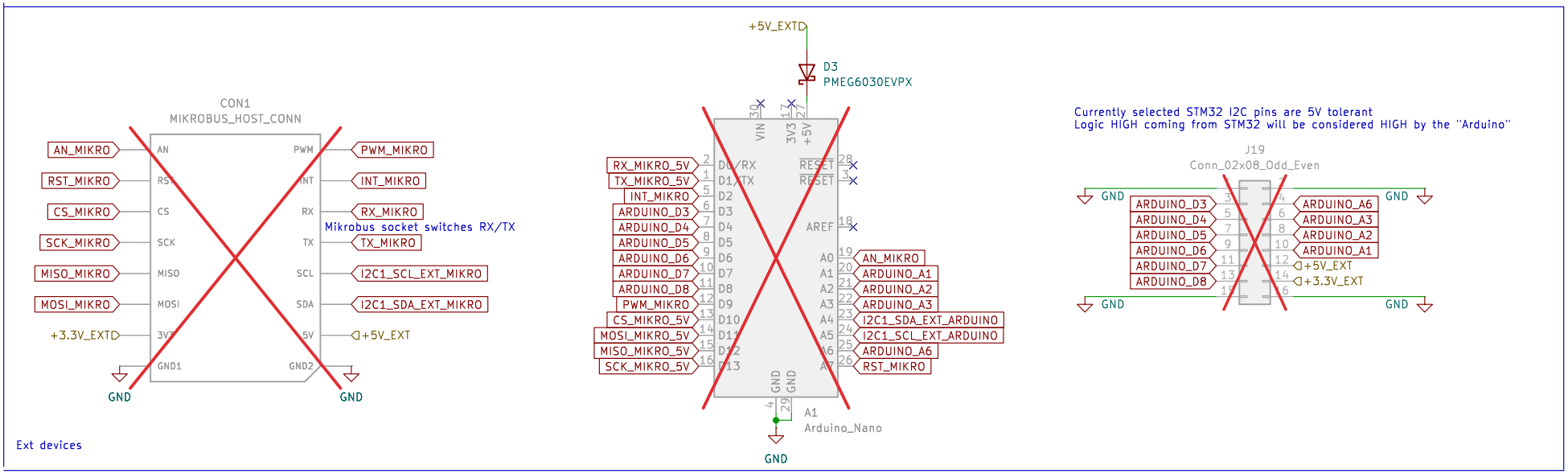
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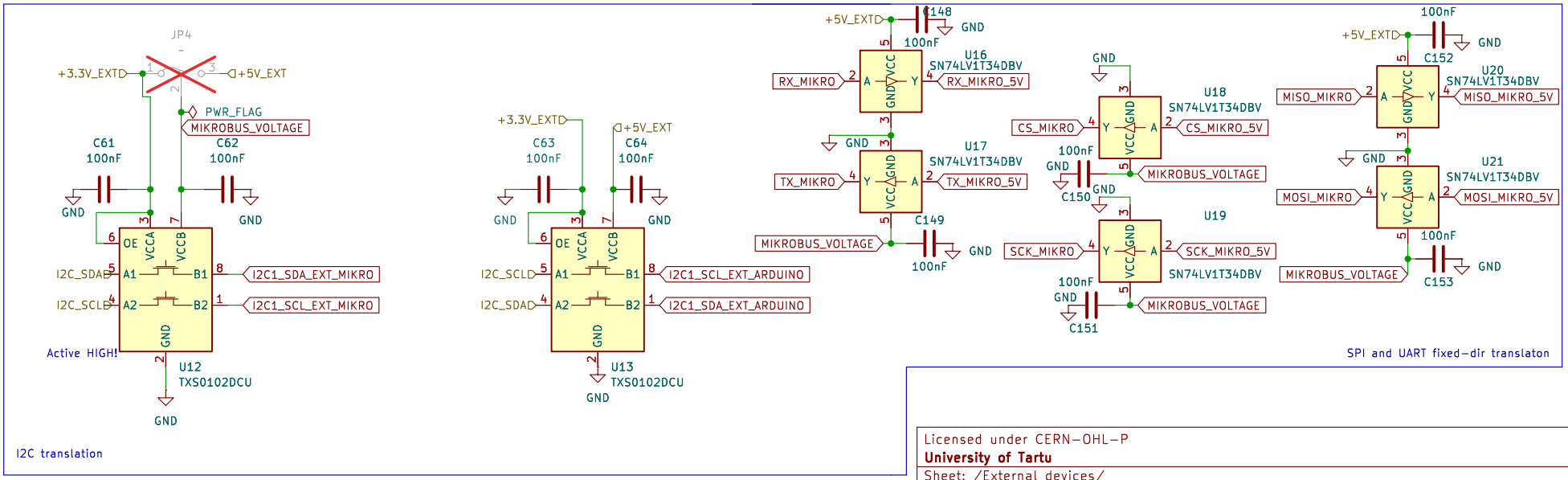
University of Tartu

Sheet: /DC-DC 12V for Motors/
File: dc_dc_12v_motors.kicad_sch

Title: DC-DC converter 12V for motors

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 8/18





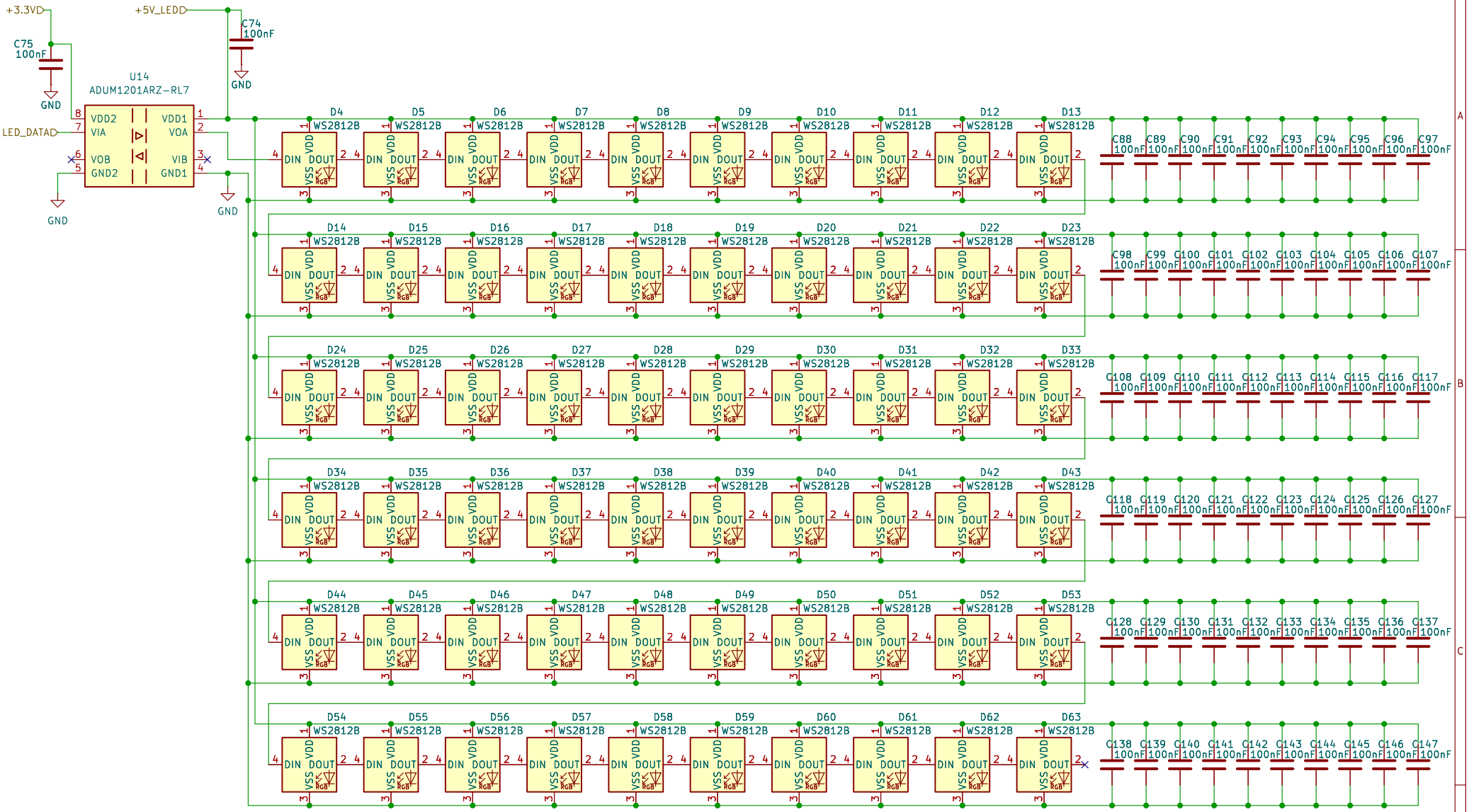
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Sheet: /External devices/
File: ext_devices.kicad_sch

Title: External devices

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 9/18



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Sheet: /Addressable LEDs around the PCB/
File: leds.kicad_sch

Title: LEDs around the mainboard

Size: A4	Date: 2023-21-11	**Rev: 0.2.0**
KiCad E.D.A. 9.0.6		Id: 10/18